

Dr. Dora Rajonhson

Computational Biologist · Machine Learning & Bioinformatics · Multi-omics
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PROFESSIONAL PROFILE

PhD-trained computational biologist with 8 years of research experience applying machine learning and statistical modelling to genomic, transcriptomic, and multi-omics datasets. Proven ability to build scalable bioinformatics pipelines from raw NGS reads to interpretable biological results across radically different biological systems. Expert in RNA-seq, metagenomics, 16S rRNA, and integrated omics analysis. Combines rigorous computational skills with deep biological intuition. First-author publications in peer-reviewed journals; track record of independent project delivery in HPC environments.

TECHNICAL SKILLS

Languages	R (DESeq2, edgeR, Bioconductor, tidyverse), Python (Biopython, pandas, NumPy, matplotlib), Linux / bash / shell
ML & Statistics	Supervised classification: Random Forest (randomForest, R) for taxon-level feature importance in disease resistance modelling; ANN-based classification (XYOM); discriminant analysis. GLM dose-response modelling (logit-link, R). Unsupervised dimensionality reduction (PCA, NMDS), PERMANOVA, Sloan neutral model
Transcriptomics	Bulk RNA-seq end-to-end: QC (FastQC/MultiQC), trimming, alignment (STAR, HISAT2), quantification, DESeq2 / edgeR, GO/KEGG enrichment; multi-omics integration
Genomics & Metagenomics	16S rRNA (DADA2, QIIME2, short-read & full-length), shotgun metagenomics (Kraken2/Bracken), viromics, AMR profiling, community-level statistics
Pipelines	HPC (SLURM), reproducible workflow design, large-scale NGS data management
Bioinformatics Tools	STAR, HISAT2, DESeq2, QIIME2/DADA2, Kraken2/Bracken, PICRUST2, fastp, BWA, samtools, phyloseq, Lefse

RESEARCH & PROFESSIONAL EXPERIENCE

Industrial Postdoctoral Fellow - Multi-omics & Food Safety 2024-2025

BIOTEC - National Center for Genetic Engineering and Biotechnology, Thailand | PMUC Fellowship

- Built and ran 16S rRNA sequencing pipelines for longitudinal microbiome profiling in closed aquaculture systems across 13-week production cycles.
- Led the first multi-mycotoxin transcriptomic study in shrimp (*Penaeus. monodon*): end-to-end bulk RNA-seq (Illumina NovaSeq) from QC and normalisation through DESeq2 differential expression, GO/KEGG pathway, and parallel 16S rRNA microbiome profiling on intestine and hepatopancreas.
- Applied longitudinal modelling and environmental source tracking to trace microbial transmission between water and host gut; produced reproducible pipelines and documented methodologies for consortium reporting.
- Supervised junior researchers and provided technical guidance on sequencing preparation, data management, and bioinformatics analysis.

Postdoctoral Fellow - Microbiome Surveillance & Bioinformatics 2023-2024

BIOTEC - UNESCO/TWAS Fellowship, Thailand

- Benchmarked Illumina short-read vs. Oxford Nanopore full-length 16S rRNA sequencing strategies: systematic comparison of taxonomic resolution, cost-effectiveness, and analytical performance across community-level metrics.
- Evaluated probiotic interventions using Spearman correlation-based microbial networks (igraph, Cytoscape) and PICRUST2 functional pathway prediction.
- Applied Random Forest (randomForest, R) to identify microbial taxa associated with White Spot Syndrome Virus (WSSV) resistance and susceptibility in shrimp gut microbiota, using feature importance scores to extract key community signals from a multi-study dataset.

PhD Research Fellow - Medical Entomology & Zoonotic Disease Surveillance 2017-2022

Mahidol University - OWSD-UNESCO/SIDA Fellowship, Thailand

- Produced the first metagenomic and viromic characterisation of bat bug (*Stricticimex parvus*) and swallow bug (*Paracimex avium*) microbiomes: sequenced DNA and RNA fractions across two species and three life stages.
- Applied taxonomic profiling (Kraken2/Bracken), alpha/beta diversity analysis, and multivariate statistics to identify community structure drivers, flag zoonotic taxa, and screen for AMR genes; RNA fraction enabled RNA virus detection extending surveys beyond bacterial communities.
- Applied geometric morphometrics and linear morphometry with ANN-based classification for species identification in Cimicidae (bed bugs, bat bugs, swallow bugs) comparing traditional and outline-based approaches for supervised taxonomic discrimination.
- Investigated Cell Fusing Agent Virus (CFAV) as a microbiome-based biological control candidate for dengue virus.

MSc Researcher - Plague Vector Control & Predictive Modelling

2014-2015

Institut Pasteur de Madagascar - Gérard Scholarship

- Evaluated fipronil as a systemic insecticide against *Xenopsylla cheopis*, the primary plague flea vector in Madagascar, across three experimental delivery methods: artificial blood-feeding device, live rodent hosts (*Rattus norvegicus* and *R. rattus*), and WHO contact bioassay.
- Built dose-response predictive models using generalised linear models (GLM, binomial distribution, logit-link) in R to compute LD50 and LD90 values across experimental conditions

EDUCATION

PhD in Tropical Medicine

2017-2022

Mahidol University, Thailand | Most Outstanding PhD Student - Tranakchit Harinasuta Award (2018)

MSc in Biology

2013-2015

University of Antananarivo, Madagascar

TEACHING & SUPERVISION

Adjunct Lecturer & PhD Co-Supervisor

Dec 2025-present

University of Antananarivo, Madagascar

Co-supervising a PhD student on Kdr gene amplicon sequencing and phylogeographic analysis of plague vectors.

MSc Supervisor

2024-2025

Thammasat University, Thailand

Supervised MSc student on microbial community profiling; mentored in bioinformatics pipeline development and scientific writing.

PUBLICATIONS

Published (peer-reviewed, first-author):

- Rajonhson DM, Angthong P, Thepsuwan T, et al. Integrating short- and full-length 16S rRNA gene sequencing to elucidate microbiome profiles in Pacific white shrimp (*Litopenaeus vannamei*) ponds. *Microbiology Spectrum*. 2024; 12(11). doi:10.1128/spectrum.00965-24
- Rajonhson DM, Raksakoon C, Payakkapol A, Dujardin S, Dujardin J-P, Potiwat R. Comparison of Two Different Morphological Methods to Study the Pronotum of Cimicidae: Bed Bugs, Bat Bugs, and Swallow Bugs. *Insects*. 2022; 13(12):1155. doi:10.3390/insects13121155
- Rajonhson DM, Miarinjara A, Rahelinirina S, Rajerison M, Boyer S. Effectiveness of Fipronil as a Systemic Control Agent Against *Xenopsylla cheopis* (Siphonaptera: Pulicidae) in Madagascar. *Journal of Medical Entomology*. 2017. doi:10.1093/jme/tjw200

Under review (first-author):

- Rajonhson DM et al. Microbiome dynamics in closed aquaculture systems: environmental drivers and implications for sustainable shrimp farming. *Aquaculture*.
- Rajonhson DM et al. Hidden threats in aquafeeds: multi-mycotoxin contamination and its molecular impacts on shrimp health and food safety. *Food and Chemical Toxicology*.
- Rajonhson DM et al. Ectoparasite microbiomes as sentinels of zoonotic risk: a metagenomic approach to wildlife-associated vector surveillance. *PLOS Neglected Tropical Diseases*.

Full publication list, conference presentations, and references available on request.

FELLOWSHIPS & AWARDS

- BIOTEC-PMUC Industrial Postdoctoral Fellowship (2024-2025)
- BIOTEC-UNESCO-TWAS Postdoctoral Fellowship (2023-2024)
- OWSD-UNESCO-SIDA PhD Fellowship (2017-2022)
- Gérard Scholarship for Master's Thesis, Institut Pasteur de Madagascar (2014-2015)

SELECTED ADVANCED TRAINING

- Viral Genomics and Bioinformatics - Wellcome Connecting Science / OUCRU, Vietnam (2024)
- Sustainable Resources for CRISPR-based Diagnostics - Protein Society of Thailand (2023)
- Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) - University of Washington (2020)
- International Short Course on Ecosystem Health - Thailand One Health University Network (2018)